

Greenplum

Technical Publications

Greenplum Database 3.2 Performance Monitor Guide

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Preface

This guide provides information for system administrators and Greenplum Database super users responsible for administering the Greenplum Performance Monitor.

- About This Guide
- Greenplum Database Documentation
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About This Guide

This guide provides information and instructions for installing, configuring and maintaining Greenplum Performance Monitor. This guide is intended for system administrators and database super users responsible for managing the Greenplum system and databases.

This guide assumes knowledge of Linux/Unix system administration, database management systems, database administration, and structured query language (SQL).

Because Greenplum Database is based on PostgreSQL 8.2, this guide assumes some familiarity with PostgreSQL. Links and cross-references to PostgreSQL documentation are provided throughout this guide for features that are similar to those in Greenplum Database.

Greenplum Database Documentation

The Greenplum Database documentation is provided in PDF and HTML format. The documentation can be found in the `$GPHOME/docs` directory of your Greenplum Database installation, or on the Greenplum Network.

The Greenplum Database documentation is intended as a supplement to the PostgreSQL 8.2 documentation, upon which Greenplum Database is based.

The following documents are provided with this release of Greenplum Database:

- **Release Notes** — The release notes (`README`) explain the new features and any known issues associated with this release.
- **Greenplum Database Administrator Guide** — This guide contains conceptual, instructional and reference information for database and system administrators responsible for installing, configuring and administering a Greenplum Database system.

Document Conventions

The following conventions are used throughout the Greenplum Database documentation to help you identify certain types of information.

Text Conventions

Table 0.1 Text Conventions

Text Convention	Usage	Examples
bold	Button, menu, tab, page, and field names in GUI applications	Click **Cancel** to exit the page without saving your changes.
italics	New terms where they are defined Database objects, such as schema, table, or columns names	The *master instance* is the postmaster process that accepts client connections. Catalog information for Greenplum Database resides in the *pg_catalog* schema.
`monospace`	File names and path names Programs and executables Command names and syntax Parameter names	Edit the `postgresql.conf` file. Use `gpstart` to start Greenplum Database.
`monospace italics`	Variable information within file paths and file names Variable information within command syntax	`/home/gpadmin/`*`config_file`* `COPY` *`tablename`* `FROM '`*`filename`*`'`
`monospace bold`	Used to call attention to a particular part of a command, parameter, or code snippet.	Change the host name, port, and database name in the JDBC connection URL: `jdbc:postgresql://`**`host:5432/mydb`**
`UPPERCASE`	Environment variables SQL commands Keyboard keys	Make sure that the Java `/bin` directory is in your `$PATH`. `SELECT * FROM my_table;` Press `CTRL+C` to escape.

Command Syntax Conventions

Table 0.2 Command Syntax Conventions

Text Convention	Usage	Examples
`{ }`	Within command syntax, curly braces group related command options. Do not type the curly braces.	`FROM { 'filename' | STDIN }`
`[ ]`	Within command syntax, square brackets denote optional arguments. Do not type the brackets.	`TRUNCATE [ TABLE ] name`
`...`	Within command syntax, an ellipsis denotes repetition of a command, variable, or option. Do not type the ellipsis.	`DROP TABLE name [, ...]`
`|`	Within command syntax, the pipe symbol denotes an "OR" relationship. Do not type the pipe symbol.	`VACUUM [ FULL | FREEZE ]`

`$ system_command`
`# root_system_command`
`=> gpdb_command`
`=# su_gpdb_command`
	Denotes a command prompt - do not type the prompt symbol. `$` and `#` denote terminal command prompts. `=>` and `=#` denote Greenplum Database interactive program command prompts (`psql` or `gpssh`, for example).	`$ createdb mydatabase`
		`# chown gpadmin -R /datadir`
		`=> SELECT * FROM mytable;`
		`=# SELECT * FROM pg_database;`

Naming Conventions and Acronyms

Table 0.3 Naming Conventions and Acronyms

Convention	Meaning
`$GPHOME`	The base directory where Greenplum Database is installed, for example: `/usr/local/greenplum-db`
`gpadmin`	The default name for the Greenplum Database super user.
`KB`	Kilobytes (2^{10})
`MB`	Megabytes (2^{20})
`GB`	Gigabytes (2^{30})
`TB`	Terabytes (2^{40})

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support@greenplum.com

1 About Greenplum Performance Monitor

Greenplum provides an optional performance monitoring feature that administrators can install and enable for Greenplum Database 3.2. Greenplum Performance Monitor lets administrators view the status and performance of actively running queries and allows for detailed analysis of historical queries and system metrics. A browser-based graphical interface provides easy access to a variety of configurable views and reports on performance data.

Greenplum Performance Monitor Architecture

Greenplum Performance Monitor is based on an agent architecture. Agent processes are installed on each segment host, and all other monitor components are installed on the master host.

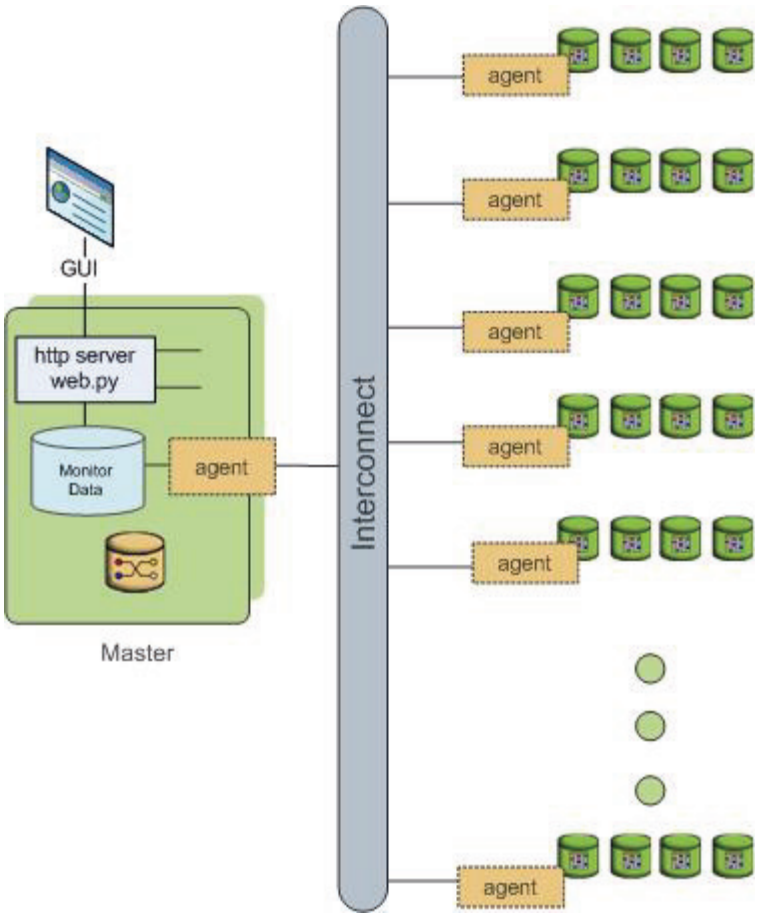


Figure 1.1 Greenplum Performance Monitor Architecture

The agents collect performance data about active queries and system utilization and send the data to the Greenplum master at regular intervals. The data is stored in a dedicated Greenplum Performance Monitor database on the master, where it can be accessed through the user interface or through SQL queries.

Greenplum Performance Monitor Agents

Greenplum Performance Monitor agents run on all hosts in a Greenplum Database array. Each segment host contains its own `gpsmon` agent process, which collects runtime statistics for system metrics, queries, and iterators. A master agent `gpmmon` process controls and maintains connections to the segment agents.

At runtime, segment agent processes send performance data to the master agent at configurable intervals called `quantum`. The master agent pulls together the data from all agents, stores it in flat files, and periodically commits the data in the files to the Greenplum Performance Monitor database.

Greenplum Performance Monitor Database

The Greenplum Performance Monitor database is a distributed Greenplum database dedicated to storing and serving performance data for the system. The `gpperfmon` installer runs SQL cripts to build and initialize this database on the master host.

The database consists of two sets of tables, one set for current data and another set for historical data.

Table 1.1 Greenplum Performance Monitor Data Tables

Current Data Table (External)	Historical Data Table
metrics_now	*metrics_history*
database_now	*database_history*
queries_now	*queries_history*
iterators_now	*iterators_history*

The current data is stored in flat files that are treated as external tables by the database. These files are committed to historical tables at an interval of three times the value of `quantum`.

The history tables are partitioned by month, allowing you to drop partitions for data that you no longer need. The `gpperfmon` installer creates eighteen months of partitioned tables. For more information on these tables and the Greenplum Performance Monitor database schema, see Appendix B, “Greenplum Performance Monitor Catalog Reference”.

Greenplum administrators can connect to the monitor database to manage and query performance data using client programs such as `psql` or application programming interfaces (APIs) such as JDBC and ODBC. Also, administrators can view reports on current and historical peformance data in the Greenplum Performance Monitor graphical interface.

Greenplum Performance Monitor User Interface

Greenplum Performance Monitor provides a graphical interface for viewing performance metrics. With this browser-based application, you can create customized graphical views for tracking the performance of active queries and for analyzing historical data for queries and system metrics.

<TBD: screen shot of GUI with callouts>

For more information, see Chapter 4, “Using Greenplum Performance Monitor”.

Greenplum Performance Monitor Web Server

The Greenplum Performance Monitor user interface is powered by a lightweight lighttpd web server. The lighttpd server is an open-source web server with a low memory footprint and light CPU load. For more information, see http://www.lighttpd.net/.

The `gpperfmon` installer sets up the lighttpd web server, prompting you for basic configuration information on the desired port and SSL options. Under normal conditions, the web server requires minimal maintenance and administration, as described in “Web Server Administration” on page 17.

<TBD: some sort of “follow a query through the monitor” example or graphic? Would it go here or up with the architecture diagram? Here, it would restate the overall process in context of the more detailed discussion of each component>

2 Installing and Configuring Greenplum Performance Monitor

You must be the Greenplum superuser (`gpadmin`) to install Greenplum Performance Monitor. Installing and configuring Greenplum Performance Monitor involves the following high-level steps:

- Preparing for Installation
- Installing the Segment Monitor Agents
- Installing Greenplum Performance Monitor on the Master
- Configuring Greenplum Performance Monitor

This section also provides information about reconfiguring and uninstalling Greenplum Performance Monitor.

Preparing for Installation

Before installing Greenplum Performance Monitor, verify that your system meets the basic requirements, create a monitor superuser, and change settings in `postgresql.conf`.

System Requirements

<TBD: How much of the master's RAM, CPU and disk will be used by the monitor and web server; get values from QA>

Creating the Monitor Superuser

The Greenplum Performance Monitor requires a dedicated superuser role with database connection privileges. The monitor superuser owns the database that stores performance statistics data, connects to segment hosts to control agents, and performs other monitor-related operations.

To create the gpmmon superuser and assign connection privileges:

1. Add a new user with the name `gpmon`. For example:

   ```
   # useradd gpmon
   ```

2. Using `psql` or another PostgreSQL-compatible client program, create a new superuser role with the `CREATEDB` attribute. For example:

   ```
   $ psql template1 -c 'CREATE ROLE gpmon WITH SUPERUSER
   CREATEDB LOGIN;'
   ```

3. Edit `$MASTER_DATA_DIRECTORY/pg_hba.conf` to give the `gpmmon` superuser privileges to connect to the database. For example, to allow trusted local connections only, add the following entries to the file:

   ```
   local     all     gpmon     trust
   ```

```
host      all     gpmon     0.0.0.0/0 trust
```

4. Use `gpstop` with the `-u` option to reload `pg_hba.conf`:

   ```
   $ gpstop -u
   ```

Editing postgresql.conf

To enable the monitor, you must change and verify settings as described in this section.

To configure Greenplum Performance Monitor:

1. Edit `$MASTER_DATA_DIRECTORY/postgresql.conf` and set the following parameter to "on" as shown<TBD at time of writing, this must be added>:

   ```
   gp_enable_gpperfmon=on
   ```

2. Find the following parameter in the "EXTERNAL TABLES" section of `postgresql.conf` and verify that it is set to "on" as shown:

   ```
   gp_external_enable_exec=on
   ```

3. If you want to use a different port than the default port 8888, set the following parameter:

   ```
   gpmon_port=<port_number>
   ```

4. Use `gpstop` with the `-u` option to reload `postgresql.conf`:

   ```
   $ gpstop -u
   ```

Installing the Segment Monitor Agents

<TBD: this is untested at time of writing>

To install Monitor Agents on the Segment Hosts

1. On the master host, create a tar file of your Greenplum Performance Monitor installation. For example (running as `root`):

   ```
   # su -
   # cd /usr/local
   # gtar -cvf /home/gpadmin/gp.tar <TBD: package name?>
   ```

2. Copy the tar file to the segment hosts using `gpscp`. For example:

   ```
   # source /usr/local/greenplum-db/greenplum_path.sh
   # gpscp -f /home/gpadmin/single_seg_hosts_file
   /home/gpadmin/gp.tar =:/usr/local
   ```

3. Start an interactive session in `gpssh`. For example:

   ```
   # gpssh -f /home/gpadmin/single_seg_hosts_file
   ```

4. At the `gpssh` command prompt, untar the tar file in the installation directory on the segment hosts. For example:

   ```
   => gtar --directory /usr/local -xvf /usr/local/gp.tar
   ```

5. Confirm that the Greenplum Performance Monitor directory was installed in the correct location (the same location as `gpperfmon` on your master host). For example:

   ```
   => ls /usr/local/greenplum-perfmon-2.0.0
   ```

6. Change the ownership of the Greenplum Database install directory to the `gpadmin` user or group. For example:

   ```
   => chown -R gpadmin /usr/local/greenplum-perfmon-2.0.0
   => chgrp -R gpadmin /usr/local/greenplum-perfmon-2.0.0
   ```

7. Remove the tar file. For example:

   ```
   => rm /usr/local/gp.tar
   ```

Installing Greenplum Performance Monitor on the Master

This section describes how to copy the binaries for Greenplum Performance Monitor to your master host. After this installation procedure is complete, you must follow the steps described in the next section, Configuring Greenplum Performance Monitor, to finish setting up the monitor.

To install Greenplum Performance Monitor

1. Download or copy the installer file to the Greenplum Database master host. Installer files are available from Greenplum for RedHat (32-bit and 64-bit) and Solaris (64-bit) platforms.

2. Unzip the installer file where *`PLATFORM`* is either `RHEL4-i386` (RedHat 32-bit), `RHEL4-x86_64` (RedHat 64-bit) or `SOL-x86_64` (Solaris 64-bit). For example:

   ```
   # unzip greenplum-perfmon-3.2.0.0-PLATFORM.zip
   ```

3. Launch the installer using `bash`. For example:

   ```
   # /bin/bash greenplum-perfmon-3.2.0.0-PLATFORM.bin
   ```

4. The installer will prompt you to accept the Greenplum Performance Monitor license agreement. Type `yes` to accept the license agreement.

5. The installer will prompt you to provide an installation path. Press `ENTER` to accept the default install path (`/usr/local/greenplum-perfmon-2.0.0`), or enter an absolute path to an install location. You must have write permissions to the location you specify.

   When you see the following message, you have successfully installed Greenplum Performance Monitor, and can continue to Configuring Greenplum Performance Monitor:

```
**************************************************************

Installation complete.

Greenplum Database Performance Monitor is installed in
<installation_dir>.

Please read <installation_dir>/docs/README to get started.
```

****

Configuring Greenplum Performance Monitor

The `gpperfmon` utility sets up and configures Greenplum Performance Monitor on the master. This utility installs the following components on your master host:

- Dedicated Greenplum database for performance data (`gpperfmon`)
- Master monitor agent (`gpmmon`)
- HTTP server and scripts to support the user interface

During the configuration process, the utility prompts you to configure these components, and accordingly edits `gpperfmon.conf` and the web server configuration file, `lighttpd.conf`. The following table lists the required and optional configuration values in the order that they are prompted during installation.

Table 2.1 Checklist for Greenplum Performance Monitor configuration

Parameter	Default Value
`quantum`	15 seconds
`min_query_time`	60 seconds
`min_detailed_query_time`	Query run time value
`log_location`	$MASTER_DATA_DIRECTORY/gpperfmon/logs
`max_data_points`	600
Thresholds for UI alerts	
`load2_threshold`	2.00
`cpu_threshold`	80
`memory_threshold`	80
`disk_threshold`	0
`network_threshold`	10000
`cpu_elapsed_threshold`	70
`skew_cpu_threshold`	100
Web server port	8080
Optional SSL setup for connections to the user interface	
Country Name (See country codes)	
State or Province Name	
Locality Name	
Organization Name	
Organizational Unit Name	
Common Name	
Email Address	

For more information on choosing values for your environment, see detailed descriptions of the configuration parameters in Appendix A, "Utility and Configuration Reference: gpperfmon"

To accept the displayed default values for any of these parameters at configuration time, hit the `ENTER` key.

To configure Greenplum performance monitor

1. Launch `gpperfmon` with the `--setup` option. For example:
   ```
   $ gpperfmon --setup
   ```
2. The utility prompts you to create the Greenplum Performance Monitor database. Type `Y` to agree and then type `Y` again to verify. The installer runs SQL scripts to create the database and displays the following message:
   ```
   The gpperfmon database was created successfully.
   ```
3. Enter a value for Quantum. The default is 15 seconds.
4. Enter a value for Minimum query run time. The default is 60.
5. Enter a value for Minimum detailed query run time. The default value displayed is the value you specified for Minimum query run time.
6. Enter a location for the Greenplum Performance Monitor log files. The default is `$MASTER_DATA_DIRECTORY/gpperfmon/logs`.
7. Enter a value for the maximum number of data points the Web UI can return in one call. The default is 600.
8. The monitoring GUI can highlight specific query or system metrics that exceed a certain threshold. Enter desired values for the following metrics:
   ```
   Load2 threshold [2.00]:
   CPU usage threshold (percent) [80]:
   Memory usage threshold (percent) [80]:
   Disk usage threshold (bytes) [0]:
   Network usage threshold (bytes) [10000]:
   CPU elapsed threshold [70]:
   CPU skew threshold [100]:
   ```
9. Enter the web server port for the user interface. The default 8080.
10. (Optional) Enter `Y` or `N` to set up SSL connections for the user interface. If you enter `Y`, you are prompted for the following DN information:
    ```
    Country Name (2 letter code) [GB]:
    State or Province Name (full name) [Berkshire]:
    Locality Name (eg, city) [Newbury]:, s
    Organization Name (eg, company) [My Company Ltd]:
    Organizational Unit Name (eg, section) []:
    Common Name (eg, your name or your server's hostname) []:
    ```

```
Email Address []:
```

When the utility displays the following message, you have successfully completed Greenplum Performance Monitor installation:

```
gpperfmon configuration is now complete.  If at a later date
you want to change certain parameters, you can either re-run
'gpperfmon --setup' or edit the configuration file located
at $GPMON_HOME/conf/gpperfmon.conf. You can start monitoring
this Greenplum Database instance by running 'gpperfmon
--start'.  The web based UI is available at <hostname>:8080/
```

For more information on next steps, see Starting and Stopping Greenplum Performance Monitor.

Reconfiguring Greenplum Performance Monitor

The configuration for Greenplum Performance Monitor is stored in `$MASTER_DATA_DIRECTORY/gpperfmon/conf/gpperfmon.conf`. This configuration file contains all the configuration options described in Appendix A, "Utility and Configuration Reference: gpperfmon".

If you want to change selected configuration values, edit `gpperfmon.conf`. If you want to change most or all configuration values for a monitor instance, you can run `gpperfmon -- setup`. When directly editing `gpperfmon.conf`, you must restart the monitor using `gpperfmon --restart` in order to activate the new configuration values.

Editing the Configuration File

A typical case for editing `gpperfmon.conf` is adjustment of the value for `min_detailed_query_time`. As you evaluate the amount of data being saved to the monitor database, you may want to increase the value of this parameter but otherwise leave all other configuration values unchanged.

To edit the configuration file

1. In a text editor, open `$MASTER_DATA_DIRECTORY/gpperfmon/conf/gpperfmon.conf`.
2. Locate the desired parameter and type the new value. For example, to change `min_detailed_query_time`, modify this setting found in the `[GPMMON]` section of the file:

   ```
   min_detailed_query_time=360
   ```

3. When you are finished making changes, save `gpperfmon.conf`.
4. Restart the monitor. For example:

   ```
   $ gpperfmon --restart
   ```

Reconfiguring a Monitor Instance with gpperfmon

Running `gpperfmon --setup` effectively resets all values in `gpperfmon.conf` and is useful in situations where you need to change many or all of the values for a monitor instance. Also, with `gpperfmon --setup` you can replace the existing monitor database with a fresh instance, or change values that are not available in `gpperfmon.conf`, such as the web server port and the DN values used for SSL.

To reconfigure a monitor instance with gpperfmon

1. Run `gpperfmon --setup`.
2. The utility displays the message:

   ```
   gpperfmon.conf already exists for this database instance. Do
   you want to reconfigure gpperfmon? [N]:
   ```

   Type `Y` to reconfigure the monitor instance.
3. The utility prompts you to create a new database, warning you that all existing performance data will be lost. Type `Y` or `N` as desired.
4. Enter a value for Quantum. The default is 15 seconds.
5. Enter a value for Minimum query run time. The default is 60.
6. Enter a value for Minimum detailed query run time. The default value displayed is the value you specified for Minimum query run time.
7. Enter a location for the Greenplum Performance Monitor log files. The default is `$MASTER_DATA_DIRECTORY/gpperfmon/logs`.
8. Enter a value for the maximum number of data points the Web UI can return in one call. The default is 600.
9. The monitoring GUI can highlight specific query or system metrics that exceed a certain threshold. Enter desired values for the following metrics:

   ```
   Load2 threshold [2.00]:
   CPU usage threshold (percent) [80]:
   Memory usage threshold (percent) [80]:
   Disk usage threshold (bytes) [0]:
   Network usage threshold (bytes) [10000]:
   CPU elapsed threshold [70]:
   CPU skew threshold [100]:
   ```

10. Enter the web server port for the user interface. The default 8080.
11. (Optional) Enter `Y` or `N` to set up SSL connections for the user interface. If you enter `Y`, you are prompted for the following DN information:

    ```
    Country Name (2 letter code) [GB]:
    State or Province Name (full name) [Berkshire]:
    Locality Name (eg, city) [Newbury]:, s
    Organization Name (eg, company) [My Company Ltd]:
    ```

```
Organizational Unit Name (eg, section) []:
Common Name (eg, your name or your server's hostname) []:
Email Address []:
```

When the utility displays the following message, you have successfully reconfigured Greenplum Performance Monitor:

```
gpperfmon configuration is now complete.  If at a later date
you want to change certain parameters, you can either re-run
'gpperfmon --setup' or edit the configuration file located
at $GPMON_HOME/conf/gpperfmon.conf. You can start monitoring
this Greenplum Database instance by running 'gpperfmon
--start'.  The web based UI is available at <hostname>:8080/
```

Uninstalling Greenplum Performance Monitor

Removing a Greenplum Performance Monitor instance is a simple manual process. To uninstall, you must remove monitor directories, drop the monitor database, and disable the monitor in your Greenplum Database configuration.

To uninstall Greenplum Performance Monitor

1. On the Greenplum Database master, open `postgresql.conf` and change the value of `gp_enable_gpperfmon` to `off`.
2. Reload the configuration file using the `gpstop` utility. For example:

   ```
   $gpstop -u
   ```
3. Remove the monitor installation directory. For example:

   ```
   $ rm -rf gpperfmon-3.2
   ```
4. Remove the directory that contains monitor data and configuration information. In a default setup this is `$MASTER_DATA_DIRECTORY/gpperfmon`. For example:

   ```
   $ rm -rf gpperfmon
   ```
5. Drop the monitor database using the `dropdb` utility. For example:

   ```
   $ dropdb --echo gpperfmon
   DROP DATABASE gpperfmon;
   DROP DATABASE
   ```
6. <TBD: any kind of verification step available, to make sure all the processes are dead?>

3 Administering Greenplum Performance Monitor

Much of the routine administration of Greenplum Performance Monitor is performed using the `gpperfmon` utility. This utility enables basic tasks such as stopping and restarting the monitor.

Other tasks for administration of the user interface web server or the monitor database are performed with native tools or command-line interfaces, as described in this chapter.

Starting and Stopping Greenplum Performance Monitor

The `gpperfmon` utility starts, stops and restarts Greenplum Performance Monitor with the following option flags:

```
--start
--stop
--restart
```

For example, to start Greenplum Performance Monitor, run the following command:

```
$gpperfmon --start
```

These command options control the master agent, the segement agent processes, and the UI web server. If you want to exclude the UI web server from these commands, add the flag `--noweb`. For example, to restart agents without restarting the web server, run the following command:

```
$gpperfmon --restart --noweb
```

In this release, the master `gpmmon` agent is not automatically restarted if it fails. Greenplum recommends setting up a cronograph job on the master host to start `gpmmon` on the every few minutes. If the agent is already running, the cron job fails. If the agent is not running, the cron job restarts it.

For more information about `gpperfmon`, see "Utility and Configuration Reference: gpperfmon" on page 20.

Agent Administration

Starting and Stopping Greenplum Performance Monitor Agents

<TBD: this will be done through gpperfmon from the master. A way to start/stop particular agents on particular hosts?>

Viewing and Maintaining Master Agent Log Files

Log messages for the master agent are written to `$MASTER_DATA_DIRECTORY/gpperfmon/logs/gpmmon.log` by default. To change the log file location, run `gpperfmon -- setup`, or edit the `log_location` parameter in `gpperfmon.conf`. For more information, see Reconfiguring Greenplum Performance Monitor.

If `verbose` is set to the default value of `0`, no messages are logged. To log information for troubleshooting or debugging, set this value to 1 or 2 for detailed messages.

At higher logging levels, the size of the log files may grow dramatically. If the log files grow to an unmanageable size, restart the monitor to clear out existing monitor log messages.

Adding and Removing Monitored Hosts

<TBD: sounds like this is automatic, with the monitor getting new hosts from the configuration table. Removal should work the same? Test.>

Greenplum Performance Monitor Database Administration

<TBD: how to connect, how to restore, reference to the reconfiguration section on creating/replacing a new database. . . >

Maintaining Historical Data

<TBD: details on vacuum, reindexing, etc.>

Viewing and Maintaining Database Log Files

<TBD: gpperfmon database logs, cycling them out, archiving>

Web Server Administration

<TBD: hold on web server details until lighty use is 100% certain>

Starting and Stopping the Web Server

Viewing and Maintaining Web Server Log Files

Web server log

web.py log

4 Using Greenplum Performance Monitor

<TBD: this chapter describes user goal-driven usage of the monitor. Will include descriptions of using the GUI to get information, and also some manual querying of the monitor database.>

Greenplum Performance Monitor user interface is supported for the following browsers with Macromedia Flash:

- Internet Explorer for Windows XP and Vista
- Mozilla Firefox for Windows and Linux
- Apple Safari for Macintosh

<TBD is this chapter the place to talk about the 'highlighting' params and any other UI config tasks?>

Monitoring the Current System Workload

<TBD: information and samples>

Monitoring an Active Query

<TBD: example queries and details; includes drilling down on iterator information>

Viewing Historical System Metrics

<TBD: views and queries.>

Viewing Historical Query Metrics

<TBD: significantly different types of analysis from the active queries?>

A. Utility and Configuration Reference: gpperfmon

This utility controls aspects of the Greenplum Performance Monitor, including installation, configuration, and starting and stopping monitor agents.

Synopsis

```
gpperfmon [--setup] [--start] [--stop] [--restart] [--status]
[--version] [--noweb]
```

Description

The `gpperfmon` utility controls all aspects of Greenplum Performance Monitor. This utility installs and configures the monitor, starts and stops it, and provides status and version information.

To update monitor configuration values such as quantum or minimum query run time,, run `gpperfmon --setup`.

In this release, the master `gpmmon` agent is not automatically restarted if it fails. Greenplum recommends setting up a cronograph job on the master host to start `gpmmon` on the every few minutes. If the agent is already running, the cron job fails. If the agent is not running, the cron job restarts it. The master `gpmmon` agent takes care of starting the segment agents.

Options

gpmmon Options

These options for the master agent process `gpmmon` are set by `gpperfmon --setup`, or can be edited directly in the monitor configuration file, `$MASTER_DATA_DIRECTORY/gpperfmon/conf/gpperfmon.conf`

`min_query_time`

Specifies the minimum query run time in seconds for statistics collection. The monitor logs all queries that run longer than this value in the `queries_history` table. For queries with shorter run times, no historical data is collected. Defaults to 60 seconds.

If you know that you want to collect data for all queries, you can set this parameter to a low value. Setting the minimum query run time to zero, however, collects data even for the numerous queries run by the monitor itself, creating a large amount of data that may not be useful.

`min_detailed_query_time`

Specifies the minimum iterator run time in seconds for statistics collection. The monitor logs all iterators that run longer than this value in the `iterators_history` table. For iterators with shorter run times, no data is collected.

Defaults to the value of `min_query_time`, and must always be equal to, or greater than, the value of **`-m`**. Setting **`-d`** higher than **`-m`** allows you to log detailed iterator data only for especially complex, long-running queries, while still logging basic query data for shorter queries.

Given the complexity and size of iterator data, you may need to adjust this parameter according to the size of data collected. If a cost-benefit evaluation determines that the *iterators_history* tables are growing to excessive size without providing useful information, you can raise the value of this parameter to log iterator detail for fewer queries.

`quantum`

Specifies the time in seconds between updates from performance monitor agents on all segments. Defaults to 15 seconds, with valid values limited to the range between 10 and 120 seconds.

If you prefer a less granular view of performance, or want to collect and analyze minimal amounts of data for system metrics, choose a higher quantum. <TBD: talk to Ben and others to see if this is bogus>. To collect data more frequently, choose a lower value.

`verbose`

Verbose mode shows progress and status messages of the performance tests as they are run. The default is `0`, which displays and logs no progress or status messages. Valid values of `1` and `2` display increasingly detailed messages, and are recommended only for troubleshooting purposes. These higher settings cause dramatic growth in log file size.

`log_location`

Specifies a directory location for monitor log files. Default is `$MASTER_DATA_DIRECTORY/gpperfmon/logs`.

User Interface Options

These options for the user interface and its web server are set by `gpperfmon --setup`. With the exception of Web Server Port, which resides in the <TBD httpd.conf probably> these values can be edited directly in the monitor configuration file, `$MASTER_DATA_DIRECTORY/gpperfmon/conf/gpperfmon.conf`.

`max_data_points`

Specifies the maximum number of data points to display in the user interface in order to prevent returning data sets of unmanageable size. Default is 600. When data sets larger than this value are requested, the data is automatically aggregated and reduced to more usable units for display in the user interface.

`Web server port`

Specifies the port for the web server that provides the user interface and web API.

load2_threshold

Sets the threshold to highlight *load2* metrics in the user interface. The *load2* column reflects the CPU load average for the fifteen-minute period prior to metric collection.

cpu_threshold

Sets the threshold to highlight CPU usage metrics in the user interface.

memory_threshold

Sets the threshold to highlight memory usage metrics in the user interface.

disk_threshold

Sets the threshold to highlight disk usage metrics in the user interface.

network_threshold

Sets the threshold to highlight network throughput metrics in the user interface.

cpu_elapsed_threshold

Sets the threshold to highlight CPU elapsed metrics in the user interface.

skew_cpu_threshold

Sets the threshold to highlight skew metrics in the user interface.

SSL Setup Options

These options for SSL connections to the user interface are set by `gpperfmon --setup`.

Country Name

Country name in format of a two-letter code. See ISO for a list of valid country codes.

State or Province Name

State or province name.

Locality Name

Locality name, such as city or town name.

Organization Name

Company name.

Organizational Unit Name

Section or function. For example, “system administration.”

Common Name

Domain name. This value requires a server name to be entered for verification purposes. Entering anything other than a domain name yields an invalid certificate.

Email Address

A valid email address.

Examples

Start the Greenplum Monitor master agent but not the web server for the user interface:

```
gppperfmon --start --noweb
```

Start the utility in order to install or to change configuration values:

```
gpperfmon --setup
```

See Also

Greenplum Performance Monitor Catalog Reference

B. Greenplum Performance Monitor Catalog Reference

This is a reference of the system catalog tables for Greenplum Performance Monitor.

The monitor database stores each performance statistic type (system metrics, queries, and iterators) in two separate tables, one for current statistics and one for historical data. Current queries and iterators are commited to the history tables when the queries are completed. Metrics are commited to the history table at a fixed interval of three times the update quantum, as specified by the `gpmmon` option `quantum`.

History tables in the monitor database are partitioned by month, with 12 partitions created when the schema is installed.

- metrics_now
- metrics_history
- database_now
- database_history
- queries_now
- queries_history
- iterators_now
- iterators_history

metrics_now

The table *metrics_now* contains current system utilization metrics. This external table resides in `$MASTER_DATA_DIRECTORY/gpperfmon/data/` . Current system utilization data is stored in *metrics_now* during the period between data collection from the monitor agents and automatic commitment to the *metrics_history* table.

Table J.1 gpmon_2.metrics_now

column	type	description
ctime	timestamp	Time this row was created.
hostname	varchar(64)	Hostname of the system to which the metric belongs.
mem_total	bigint	Total system memory for the system to which the metric belongs, including actual memory and swap space.
mem_used	bigint	Used system memory for the system to which the metric belongs.
mem_actual_used	bigint	Used actual memory (memory not including portion reserved for cache and buffers).
mem_actual_free	bigint	Free actual memory (memory not including portion reserved for cache and buffers).
swap_total	bigint	Total swap space for the system to which the metric belongs.
swap_used	bigint	Used swap space for the system to which the metric belongs.
swap_page_in	bigint	Number of swap pages in.
swap_page_out	bigint	Number of swap pages out.
cpu_user	float	CPU usage by the user.
cpu_sys	float	CPU usage for the system.
cpu_idle	float	Idle CPU capacity at metric collection time.
load0	float	CPU load average for the prior one-minute period.
load1		CPU load average for the prior five-minute period.
load2		CPU load average for the prior fifteen-minute period.
quantum	int	Interval between metric collection for this metric entry.
disk_ro_rate	bigint	Disk read operations per second.
disk_wo_rate	bigint	Disk write operations per second.
disk_rb_rate	bigint	Bytes per second for disk read operations.
disk_wb_rate	bigint	Bytes per second for disk write operations.
net_rp_rate	bigint	Packets per second on the system network for read operations.
net_wp_rate	bigint	Packets per second on the system network for write operations.

Table J.1 gpmon_2.metrics_now

column	type	description
net_rb_rate	bigint	Bytes per second on the system network for read operations.
net_wb_rate	bigint	Bytes per second on the system network for write operations.

metrics_history

The table *metrics_history* stores historical system utilization metrics. Like other history tables in the monitor database, it is partitioned by month, allowing you to drop partitions for data that you no longer need.

Table J.1 gpmon_2.metrics_history

column	type	description
ctime	timestamp	Time this row was created.
hostname	varchar (64)	Hostname of the system to which the metric belongs.
mem_total	bigint	Total system memory for the system to which the metric belongs.
mem_used	bigint	Used system memory for the system to which the metric belongs.
mem_actual_used	bigint	Used actual memory (memory not including portion reserved for cache and buffers).
mem_actual_free	bigint	Free actual memory (memory not including portion reserved for cache and buffers).
swap_total	bigint	Total swap space for the system to which the metric belongs.
swap_used	bigint	Used swap space for the system to which the metric belongs.
swap_page_in	bigint	Number of swap pages in.
swap_page_out	bigint	Number of swap pages out.
cpu_user	float	CPU usage by the user.
cpu_sys	float	CPU usage for the system.
cpu_idle	float	Idle CPU capacity at metric collection time.
load0	float	CPU load average for the prior one-minute period.
load1	float	CPU load average for the prior five-minute period.
load2	float	CPU load average for the prior fifteen-minute period.
quantum	int	Interval between metric collection for this metric entry.
disk_ro_rate	bigint	Disk read operations per second.
disk_wo_rate	bigint	Disk write operations per second.
disk_rb_rate	bigint	Bytes per second for disk read operations.
disk_wb_rate	bigint	Bytes per second for disk write operations.
net_rp_rate	bigint	Packets per second on the system network for read operations.
net_wp_rate	bigint	Packets per second on the system network for write operations.

Table J.1 gpmon_2.metrics_history

column	type	description
net_rb_rate	bigint	Bytes per second on the system network for read operations.
net_wb_rate	bigint	Bytes per second on the system network for write operations.

database_now

The table *database_now* contains database-wide query information for current queries. This external table resides in `$MASTER_DATA_DIRECTORY/gpperfmon/data/`. Current database-wide query data is stored in *quereies_now* during the period between data collection from the monitor agents and automatic commitment to the *database_history* table..

Table J.1 gpmon_2.iterators_history

column	type	description
ctime	timestamp	Time this row was created.
queries_total	int	The total number of queries.
queries_running	int	The number of queries currently running.
queries_queued	int	The number of queries queued.

database_history

The table *database_history* contains historical database-wide query information. Like other history tables in the monitor database, it is partitioned by month, allowing you to drop partitions for data that you no longer need..

Table J.1 gpmon_2.iterators_history

column	type	description
ctime	timestamp	Time this row was created.
queries_total	int	The total number of queries.
queries_running	int	The number of queries running at data collection time for this row.
queries_queued	int	The number of queries queued at data collection time for this row.

queries_now

The table *queries_now* contains currently running query status information. This external table resides in `$MASTER_DATA_DIRECTORY/gpperfmon/data/` . Current query status data is stored in *quieries_now* during the period between data collection from the monitor agents and automatic commitment to the *queries_history* table.

These *tmid*, *ssid* and *ccnt* columns are the composite key that uniquely identifies a query. These columns can be used to join the *queries_now* and *queries_history* tables.

Table J.1 gpmon_2.queries_now

column	type	description
ctime	timestamp	Time this row was created.
tmid	int	A time identifier for a particular query. All records associated with the query will have the same `tmid`.
ssid	int	The session id as shown by gp_session_id. All records associated with the query will have the same `ssid`.
ccnt	in	The command number within this session as shown by gp_command_count. All records associated with the query will have the same `ccnt`.
uname	varchar(64)	Username that issued this query.
db	varchar(64)	Name of the database queried.
tsubmit	timestamp	Time the query was submitted.
tstart	timestamp	Time the query was started.
tfin	timestamp	Time the query finished. For running queries, this value is always zero.
status	varchar(64)	Status of the query -- `start`, `done`, or `abort`.
rows_out	bigin	Rows out for the query.
cpu_elapsed	bigin	CPU usage for this query across all segments.
cpu_currpct	float	Current CPU percent average for all processes executing this query.
skew_cpu	float	Coefficient of variation for `cpu_elapsed` of iterators across all segments for this query, multiplied by 100. For example, a value of .95 is rendered as 95.
skew_rows	float	Coefficient of variation for `rows_in` of iterators across all segments for this query, multiplied by 100. For example, a value of .95 is rendered as 95.
query_text	text	Query text for this query.
query_plan_text	text	Query plan text for this query.

queries_history

The table *queries_history* stores historical query list status. Like other history tables in the monitor database, it is partitioned by month, allowing you to drop partitions for data that you no longer need.

These *tmid*, *ssid* and *ccnt* columns are the composite key that uniquely identifies a query. These columns can be used to join the *queries_now* and *queries_history* tables.

Table J.1 gpmon_2.queries_history

column	type	description
ctime	timestamp	Time this row was created.
tmid	int	A time identifier for a particular query. All records associated with the query will have the same `tmid`.
ssid	int	The session id as shown by gp_session_id. All records associated with the query will have the same `ssid`.
ccnt	in	The command number within this session as shown by gp_command_count. All records associated with the query will have the same `ccnt`.
uname	varchar(64)	Username that issued this query.
db	varchar(64)	Name of the database queried.
tsubmit	timestamp	Time the query was submitted.
tstart	timestamp	Time the query was started.
tfin	timestamp	Time the query finished.
status	varchar(64)	Status of the query -- `start`, `done`, or `abort`.
rows_out	bigin	Rows out for the query.
cpu_elapsed	bigin	CPU usage for this query across all segments.
cpu_currpct	float	Current CPU percent average is always zero in historical data.
skew_cpu	float	Coefficient of variation for `cpu_elapsed` of iterators across all segments for this query, multiplied by 100. For example, a value of .95 is rendered as 95.
skew_rows	float	Coefficient of variation for `rows_in` of iterators across all segments for this query, multiplied by 100. For example, a value of .95 is rendered as 95.
query_text	text	Query text for this query.

iterators_now

The table *iterators_now* contains current query iterators and their metrics. This external table resides in `$MASTER_DATA_DIRECTORY/gpperfmon/data/` . Current iterator data is stored in *iterators_now* during the period between data collection from the monitor agents and automatic commitment to the *iterators_history* table.

A query iterator refers to a node or operation in a query plan. For example, a sequential scan operation may be one node or iterator in a particular query plan. By looking at the metrics of the individual operations in a query plan, you can determine the status of a query's execution.

These *tmid*, *ssid* and *ccnt* columns are the composite key that uniquely identifies a query. These columns can be used to join the *iterators_now* and *iterators_history* tables.

Table J.1 gpmon_2.iterators_now

column	type	description
ctime	timestamp	Time this row was created.
tmid	int	A time identifier for a particular query. All iterator records associated with the query will have the same `tmid`.
ssid	int	The session id as shown by gp_session_id. All iterator records associated with the query will have the same `ssid`.
ccnt	in	The command number within this session as shown by gp_command_count. All iterator records associated with the query will have the same `ccnt`.
segid	int	The segment ID (dbid from gp_configuration).
pid	int	The process ID of the `postgres` process executing this iterator.
nid	int	The ID of this query plan node from the slice plan.
pnid	int	The ID of the parent query plan node from the slice plan.
hostname	varchar(64)	Hostname for the system where the iterator is executed.
ntype	varchar(64)	The query plan node operation type. Possible values are: `AdaptiveNestLoopJoin`, `HashAgg`, `Append`, `AppendOnlyScan`, `BitmapAnd`, `BitmapHeapScan`, `BitmapIndexScan`, `BitmapOr`, `ExternalScan`, `FunctionScan`, `Group`, `Hash`, `HashJoin`, `IndexScan`, `Limit`, `Materialize`, `MergeJoin`, `Motion`, `NestLoopJoin`, `ResultNode`, `SequentialScan`, `SetOpNode`, `ShareInputScan`, `Sort`, `SubqueryScan`, `TidScan`, `Unique`, `ValuesScan`, `Window`.
nstatus	varchar(64)	The status of this query plan node operation. Possible values are: `silent`, `submit`, `start`, `done`, or `abort`.
tstart	timestamp	Start time for the iterator.
tduration	int	Duration of the execution.
pmemsize	bigint	Memory size from Greenplum Database memory allocation data.

Table J.1 gpmon_2.iterators_now

column	type	description
pmemmax	bigint	Maximum memory from Greenplum Database memory allocation data.
memsize	bigint	Memory size from OS memory allocation data.
memresid	bigint	Resident memory, as opposed to shared or virtual memory.
memshare	bigint	Shared memory figure from OS memory allocation data.
pagefaults	bigint	Page faults for the process executing the iterator.
pagefaults_elapsed	bigint	Total page faults for the process executing the iterator.
cpu_elapsed	bigint	Total CPU usage of the process executing the iterator.
cpu_currpt	float	CPU percentage for the process executing the iterator.
phase	varchar(64)	The iterator phase.
rowsout	bigint	Number of rows output by the iterator.
rowsout_est	bigint	The query planner's estimate of rows output by the iterator.
m0_name	varchar(64)	Each operation in a query plan (ntype) has metrics associated with it. For all operations, this metric name is `Rows In`.
m0_unit	varchar(64)	The unit of measure for this metric. For all operations (ntype), this unit of measure is `Rows`.
m0_val	bigint	The value of this metric.
m0_est	bigint	The estimated value of this metric.

m1_name	varchar(64)	Each operation in a query plan (ntype) has metrics associated with it. This metric name is as follows for the corresponding plan node operations:
		`HashAgg - Agg Total Spill Tuples`
		`Append - Append Current Input Source`
		`AppendOnlyScan - AppendOnlyScan Rescan`
		`BitmapHeapScan - BitmapHeapScan Pages`
		`Hash - Hash Spill Batches`
		`HashJoin - HashJoin Spill Batches`
		`IndexScan - IndexScan Restore Pos`
		`MergeJoin - MergeJoin Inner Side Tuples`
		`NestLoopJoin - NestLoopJoin Inner Tuples`
		`SequentialScan - SeqScan Page Stats`
		`ShareInputScan - ShareInputScan Restore Pos`
		`Sort - Sort Memory Usage`
		`SubqueryScan - SubqueryScan Rescan`

Table J.1 gpmon_2.iterators_now

column	type	description

m1_unit	varchar(64)	The unit of measure for this metric. This unit is as follows for the corresponding plan node operations:
		`HashAgg - Tuples`
		`Append - Inputs`
		`AppendOnlyScan - Rescans`
		`BitmapHeapScan - Pages`
		`Hash - Batches`
		`HashJoin - Batches`
		`IndexScan - Restores`
		`MergeJoin - Tuples`
		`NestLoopJoin - Tuples`
		`SequentialScan - Pages`
		`ShareInputScan - Restores`
		`Sort - Bytes`
		`SubqueryScan - Rescans`

m1_val	bigint	The value of this metric.
m1_est	bigint	The estimated value of this metric.

m2_name	varchar(64)	Each operation in a query plan (ntype) has metrics associated with it. This metric name is as follows for the corresponding plan node operations:
		`HashAgg - Agg Total Spill Bytes`
		`Hash - Hash Spill Tuples`
		`HashJoin - HashJoin Spill Tuples`
		`IndexScan - IndexScan Rescan`
		`MergeJoin - MergeJoin Outer Side Tuples`
		`NestLoopJoin - NestLoopJoin Outer Tuples`
		`SequentialScan - SeqScan Restore Pos`
		`ShareInputScan - ShareInputScan Rescan`
		`Sort - Sort Spill Tuples`

m2_unit	varchar(64)	The unit of measure for this metric. This unit is as follows for the corresponding plan node operations:
		`HashAgg - Bytes`
		`Hash - Tuples`
		`HashJoin - Tuples`
		`IndexScan - Rescans`
		`MergeJoin - Tuples`
		`NestLoopJoin - Tuples`
		`SequentialScan - Restores`
		`ShareInputScan - Rescans`
		`Sort - Tuples`

m2_val	bigint	The value of this metric.
m2_est	bigint	The estimated value of this metric.

Table J.1 gpmon_2.iterators_now

column	type	description

m3_name	varchar(64)	Each operation in a query plan (ntype) has metrics associated with it. This metric name is as follows for the corresponding plan node operations:
		`HashAgg - Agg Total Spill Batches`
		`Hash - Hash Spill Bytes`
		`HashJoin - HashJoin Spill Bytes`
		`SequentialScan - SeqScan Rescan`
		`Sort - Sort Spill Bytes`

m3_unit	varchar(64)	The unit of measure for this metric. This unit is as follows for the corresponding plan node operations:
		`HashAgg - Batches`
		`Hash - Bytes`
		`HashJoin - Bytes`
		`SequentialScan - Rescans`
		`Sort - Bytes`

m3_val	bigint	The value of this metric.
m3_est	bigint	The estimated value of this metric.

m4_name	varchar(64)	Each operation in a query plan (ntype) has metrics associated with it. This metric name is as follows for the corresponding plan node operations:
		`HashAgg - Agg Total Spill Pass`
		`Sort - Sort Spill Pass`

m4_unit	varchar(64)	The unit of measure for this metric. This unit is as follows for the corresponding plan node operations:
		`HashAgg - Passes`
		`Sort - Passes`

m4_val	bigint	The value of this metric.
m4_est	bigint	The estimated value of this metric.

m5_name	varchar(64)	Each operation in a query plan (ntype) has metrics associated with it. This metric name is as follows for the corresponding plan node operations:
		`HashAgg - Agg Current Spill Pass Read Tuples`
		`Sort - Sort Current Spill Pass Tuples`

m5_unit	varchar(64)	The unit of measure for this metric. This unit is as follows for the corresponding plan node operations:
		`HashAgg - Tuples`
		`Sort - Tuples`

m5_val	bigint	The value of this metric.
m5_est	bigint	The estimated value of this metric.

m6_name	varchar(64)	Each operation in a query plan (ntype) has metrics associated with it. This metric name is as follows for the corresponding plan node operations:
		`HashAgg - Agg Current Spill Pass Read Bytes`
		`Sort - Sort Current Spill Pass Bytes`

Table J.1 gpmon_2.iterators_now

column	type	description

m6_unit	varchar(64)	The unit of measure for this metric. This unit is as follows for the corresponding plan node operations:
		`HashAgg - Bytes`
		`Sort - Bytes`

m6_val	bigint	The value of this metric.
m6_est	bigint	The estimated value of this metric.
m7_name	varchar(64)	Each operation in a query plan (ntype) has metrics associated with it. This metric name is as follows for the corresponding plan node operation: `HashAgg - Agg Current Spill Pass Tuples`
m7_unit	varchar(64)	The unit of measure for this metric. This unit is as follows for the corresponding plan node operation: `HashAgg - Tuples`
m7_val	bigint	The value of this metric.
m7_est	bigint	The estimated value of this metric.
m8_name	varchar(64)	Each operation in a query plan (ntype) has metrics associated with it. This metric name is as follows for the corresponding plan node operation: `HashAgg - Agg Current Spill Pass Bytes`
m8_unit	varchar(64)	The unit of measure for this metric. This unit is as follows for the corresponding plan node operation: `HashAgg - Bytes`
m8_val	bigint	The value of this metric.
m8_est	bigint	The estimated value of this metric.
m9_name	varchar(64)	Each operation in a query plan (ntype) has metrics associated with it. This metric name is as follows for the corresponding plan node operation: `HashAgg - Agg Current Spill Pass Batch`
m9_unit	varchar(64)	The unit of measure for this metric. This unit is as follows for the corresponding plan node operation: `HashAgg - Batches`
m9_val	bigint	The value of this metric.
m9_est	bigint	The estimated value of this metric.

iterators_history

The table *iterators_history* contains historical query iterators and their metrics. Like other history tables in the monitor database, it is partitioned by month, allowing you to drop partitions for data that you no longer need.

A query iterator refers to a node or operation in a query plan. For example, a sequential scan operation may be one node or iterator in a particular query plan.

These *tmid*, *ssid* and *ccnt* columns are the composite key that uniquely identifies a query. These columns can be used to join the *iterators_now* and *iterators_history* tables.

Table J.1 gpmon_2.iterators_history

column	type	description
ctime	timestamp	Time this row was created.
tmid	int	A time identifier for a particular query. All iterator records associated with the query will have the same `tmid`.
ssid	int	The session id as shown by gp_session_id. All iterator records associated with the query will have the same `ssid`.
ccnt	in	The command number within this session as shown by gp_command_count. All iterator records associated with the query will have the same `ccnt`.
segid	int	The segment ID (dbid from gp_configuration).
pid	int	The process ID of the `postgres` process executing this iterator.
nid	int	The ID of this query plan node from the slice plan.
pnid	int	The ID of the parent query plan node from the slice plan.
hostname	varchar(64)	Hostname for the system where the iterator is executed.
ntype	varchar(64)	The query plan node operation type. Possible values are: `AdaptiveNestLoopJoin`, `HashAgg`, `Append`, `AppendOnlyScan`, `BitmapAnd`, `BitmapHeapScan`, `BitmapIndexScan`, `BitmapOr`, `ExternalScan`, `FunctionScan`, `Group`, `Hash`, `HashJoin`, `IndexScan`, `Limit`, `Materialize`, `MergeJoin`, `Motion`, `NestLoopJoin`, `ResultNode`, `SequentialScan`, `SetOpNode`, `ShareInputScan`, `Sort`, `SubqueryScan`, `TidScan`, `Unique`, `ValuesScan`, `Window`.
nstatus	varchar(64)	The status of this query plan node operation. Possible values are: `silent`, `submit`, `start`, `done`, or `abort`.
tstart	timestamp	Start time for the iterator.
tduration	int	Duration of the execution.
pmemsize	bigint	Memory size from Greenplum Database memory allocation data.
pmemmax	bigint	Maximum memory from Greenplum Database memory allocation data.

Table J.1 gpmon_2.iterators_history

column	type	description
memsize	bigint	Memory size from OS memory allocation data.
memresid	bigint	Resident memory, as opposed to shared or virtual memory.
memshare	bigint	Shared memory figure from OS memory allocation data.
pagefaults	bigint	Page faults for the process executing the iterator.
pagefaults_elapsed	bigint	Total page faults for the process executing the iterator.
cpu_elapsed	bigint	Total CPU usage of the process executing the iterator.
cpu_currpt	float	CPU percentage for the process executing the iterator.
phase	varchar(64)	The iterator phase.
rowsout	bigint	Number of rows output by the iterator.
rowsout_est	bigint	The query planner's estimate of rows output by the iterator.
m0_name	varchar(64)	Each operation in a query plan (ntype) has metrics associated with it. For all operations, this metric name is `Rows In`.
m0_unit	varchar(64)	The unit of measure for this metric. For all operations (ntype), this unit of measure is `Rows`.
m0_val	bigint	The value of this metric.
m0_est	bigint	The estimated value of this metric.

m1_name	varchar(64)	Each operation in a query plan (ntype) has metrics associated with it. This metric name is as follows for the corresponding plan node operations:
		`HashAgg - Agg Total Spill Tuples`
		`Append - Append Current Input Source`
		`AppendOnlyScan - AppendOnlyScan Rescan`
		`BitmapHeapScan - BitmapHeapScan Pages`
		`Hash - Hash Spill Batches`
		`HashJoin - HashJoin Spill Batches`
		`IndexScan - IndexScan Restore Pos`
		`MergeJoin - MergeJoin Inner Side Tuples`
		`NestLoopJoin - NestLoopJoin Inner Tuples`
		`SequentialScan - SeqScan Page Stats`
		`ShareInputScan - ShareInputScan Restore Pos`
		`Sort - Sort Memory Usage`
		`SubqueryScan - SubqueryScan Rescan`

Table J.1 gpmon_2.iterators_history

column	type	description

m1_unit	varchar(64)	The unit of measure for this metric. This unit is as follows for the corresponding plan node operations:
		`HashAgg - Tuples`
		`Append - Inputs`
		`AppendOnlyScan - Rescans`
		`BitmapHeapScan - Pages`
		`Hash - Batches`
		`HashJoin - Batches`
		`IndexScan - Restores`
		`MergeJoin - Tuples`
		`NestLoopJoin - Tuples`
		`SequentialScan - Pages`
		`ShareInputScan - Restores`
		`Sort - Bytes`
		`SubqueryScan - Rescans`

m1_val	bigint	The value of this metric.
m1_est	bigint	The estimated value of this metric.

m2_name	varchar(64)	Each operation in a query plan (ntype) has metrics associated with it. This metric name is as follows for the corresponding plan node operations:
		`HashAgg - Agg Total Spill Bytes`
		`Hash - Hash Spill Tuples`
		`HashJoin - HashJoin Spill Tuples`
		`IndexScan - IndexScan Rescan`
		`MergeJoin - MergeJoin Outer Side Tuples`
		`NestLoopJoin - NestLoopJoin Outer Tuples`
		`SequentialScan - SeqScan Restore Pos`
		`ShareInputScan - ShareInputScan Rescan`
		`Sort - Sort Spill Tuples`

m2_unit	varchar(64)	The unit of measure for this metric. This unit is as follows for the corresponding plan node operations:
		`HashAgg - Bytes`
		`Hash - Tuples`
		`HashJoin - Tuples`
		`IndexScan - Rescans`
		`MergeJoin - Tuples`
		`NestLoopJoin - Tuples`
		`SequentialScan - Restores`
		`ShareInputScan - Rescans`
		`Sort - Tuples`

m2_val	bigint	The value of this metric.
m2_est	bigint	The estimated value of this metric.

Table J.1 gpmon_2.iterators_history

column	type	description

m3_name	varchar(64)	Each operation in a query plan (ntype) has metrics associated with it. This metric name is as follows for the corresponding plan node operations:
		`HashAgg - Agg Total Spill Batches`
		`Hash - Hash Spill Bytes`
		`HashJoin - HashJoin Spill Bytes`
		`SequentialScan - SeqScan Rescan`
		`Sort - Sort Spill Bytes`

m3_unit	varchar(64)	The unit of measure for this metric. This unit is as follows for the corresponding plan node operations:
		`HashAgg - Batches`
		`Hash - Bytes`
		`HashJoin - Bytes`
		`SequentialScan - Rescans`
		`Sort - Bytes`

m3_val	bigint	The value of this metric.
m3_est	bigint	The estimated value of this metric.

m4_name	varchar(64)	Each operation in a query plan (ntype) has metrics associated with it. This metric name is as follows for the corresponding plan node operations:
		`HashAgg - Agg Total Spill Pass`
		`Sort - Sort Spill Pass`

m4_unit	varchar(64)	The unit of measure for this metric. This unit is as follows for the corresponding plan node operations:
		`HashAgg - Passes`
		`Sort - Passes`

m4_val	bigint	The value of this metric.
m4_est	bigint	The estimated value of this metric.

m5_name	varchar(64)	Each operation in a query plan (ntype) has metrics associated with it. This metric name is as follows for the corresponding plan node operations:
		`HashAgg - Agg Current Spill Pass Read Tuples`
		`Sort - Sort Current Spill Pass Tuples`

m5_unit	varchar(64)	The unit of measure for this metric. This unit is as follows for the corresponding plan node operations:
		`HashAgg - Tuples`
		`Sort - Tuples`

m5_val	bigint	The value of this metric.
m5_est	bigint	The estimated value of this metric.

m6_name	varchar(64)	Each operation in a query plan (ntype) has metrics associated with it. This metric name is as follows for the corresponding plan node operations:
		`HashAgg - Agg Current Spill Pass Read Bytes`
		`Sort - Sort Current Spill Pass Bytes`

Table J.1 gpmon_2.iterators_history

column	type	description

m6_unit	varchar(64)	The unit of measure for this metric. This unit is as follows for the corresponding plan node operations:
		`HashAgg - Bytes`
		`Sort - Bytes`

m6_val	bigint	The value of this metric.
m6_est	bigint	The estimated value of this metric.
m7_name	varchar(64)	Each operation in a query plan (ntype) has metrics associated with it. This metric name is as follows for the corresponding plan node operation: `HashAgg - Agg Current Spill Pass Tuples`
m7_unit	varchar(64)	The unit of measure for this metric. This unit is as follows for the corresponding plan node operation: `HashAgg - Tuples`
m7_val	bigint	The value of this metric.
m7_est	bigint	The estimated value of this metric.
m8_name	varchar(64)	Each operation in a query plan (ntype) has metrics associated with it. This metric name is as follows for the corresponding plan node operation: `HashAgg - Agg Current Spill Pass Bytes`
m8_unit	varchar(64)	The unit of measure for this metric. This unit is as follows for the corresponding plan node operation: `HashAgg - Bytes`
m8_val	bigint	The value of this metric.
m8_est	bigint	The estimated value of this metric.
m9_name	varchar(64)	Each operation in a query plan (ntype) has metrics associated with it. This metric name is as follows for the corresponding plan node operation: `HashAgg - Agg Current Spill Pass Batch`
m9_unit	varchar(64)	The unit of measure for this metric. This unit is as follows for the corresponding plan node operation: `HashAgg - Batches`
m9_val	bigint	The value of this metric.
m9_est	bigint	The estimated value of this metric.